

AMIT KUMAR

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EDUCATION

Government Engineering College Palamu

B.Tech in Computer Science & Engineering

CGPA: 7.10 / 10 (Jharkhand University of Technology, Ranchi)

Aug. 2023 – May 2027

Palamu, JH

Inter College Domchanch

JAC Intermediate Examination

Apr. 2020 – Mar. 2022

Koderma, JH

Score: 86.8%

EXPERIENCE

Research Intern — NIT Durgapur

Biomedical Machine Learning

May 2026 – Jul. 2026

Durgapur, WB

- Developed a non-invasive anemia detection model using smartphone-based nail-bed video capture, enabling accessible screening without blood tests.
- Engineered features from capillary refill time and red-channel intensity signals, applying a hybrid ML ensemble for hemoglobin estimation.
- Worked with computer vision pipelines (OpenCV/MediaPipe) to extract and validate physiological signals from smartphone camera video.

Software Developer Intern — VS Arogya Meda Pvt. Ltd.

Healthcare Technology

Jul. 2026 – Aug. 2026 (Ongoing)

Remote/On-site

- Developing an internal billing software system for the company using Node.js, Express, and PostgreSQL.
- Building an admin web dashboard and a React Native customer-facing mobile app as part of the billing platform architecture.
- Collaborating with stakeholders to define invoicing workflows, customer records, and payment tracking features.

LEADERSHIP & ACHIEVEMENTS

College Coding Club

President

Jan. 2025 – Present

Government Engineering College Palamu

- Lead and manage a coding club of 150+ active student members, overseeing all technical and administrative operations.
- Organize coding contests, hackathons, workshops, and technical seminars to foster a culture of innovation.
- Built the official Coding Club Web Application (MERN stack) to streamline event management and registrations.
- Mentor junior students on full-stack development, DSA, and competitive programming concepts.

PROJECTS

[Coding Club Web Application](#) | React.js, Node.js, MongoDB, JWT, REST API

Mar. 2026

- Developed a full-stack web application to manage college coding club events, registrations, and announcements.
- Implemented JWT authentication and role-based access control (RBAC) for admin and student user roles.
- Designed and integrated RESTful APIs with MongoDB enabling real-time data storage and club management.
- Built responsive dashboards that significantly reduced manual coordination effort for 150+ club members.

[MeetUpX](#) | React.js, Node.js, Express.js, MongoDB, Socket.io

Jan. 2026

- Built a full-stack social media platform inspired by Instagram, letting users share photos and reels and chat with each other in real time.
- Implemented real-time one-to-one and group messaging using Socket.io, with live delivery and online-status indicators.
- Developed a Stories feature with 24-hour auto-expiry and a personalized feed showing photo and reel posts from followed users.
- Designed a mobile-responsive UI with a dedicated feed, reel viewer, and chat interface, deployed end-to-end on Render.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML5, CSS3

Frameworks: React.js, Node.js, Express.js, Tailwind CSS

Databases: MongoDB, MySQL, DBMS

AI / ML: PyTorch, NLTK, NumPy, Pandas, Matplotlib, MediaPipe, Scikit-learn

Developer Tools: Git, GitHub, VS Code, Postman, Anaconda Navigator, Jupyter Notebook

Concepts: Data Structures & Algorithms, OOP, REST API Design, Network Security, Agile

CERTIFICATIONS

- Smart India Hackathon (SIH) 2025 — Finalist, Government of India (Ministry of Education)
- Python 3.4.3 Training — Spoken Tutorial, IIT Bombay (EduPyramids / MHRD)